On the role of different annealing heat treatments on mechanical properties and microstructure of selective laser melted and conventional wrought Ti-6Al-4V

AmirMahyar Khorasani, Ian Gibson, Moshe Goldberg and Guy Littlefair
School of Engineering, Deakin University, Waurn Ponds, Australia

Abstract

Purpose – The purpose of this study was to conduct various heat treatments (HT) such as stress relief annealing, mill annealing, recrystallization ($\alpha + \beta$) annealing and β annealing followed by furnace cooling (FC) that were implemented to determine the effect of these on mechanical properties and the microstructure of selective laser melted and wrought samples. The mentioned annealings have been carried out to achieve the related standards in the fabrication of surgery implants.

Design/methodology/approach – In this paper, based on F2924-14 ASTM standard SLM and conventionally wrought parts were prepared. Then HT was performed and different characteristics such as microstructure, mechanical properties, macro-hardness and fracture surface for selective laser melted and wrought parts were analysed.

Findings – The results show that the high cooling rate in selective laser melting (SLM) generates finer grains. Therefore, tensile strength and hardness increase along with a reduction in ductility was noticed. Recrystallization annealing appears to give the best combination of ductility, strength and hardness for selective laser melted parts, whilst for equivalent wrought samples, increasing HT temperature results in reduction of mechanical properties.

Originality/value – The contributions of this paper are discussing the effect of different annealing on mechanical properties and microstructural evolution based on new ASTM standards for selective laser melted samples and comparing them with wrought parts.

Keywords Mechanical properties, Selective laser melting, Fracture surfaces, Heat treatment, Macro-hardness

Paper type Research paper

1. Introduction

Rapid prototyping and 3D printing are common names for additive manufacturing (AM) that fabricate different parts by adding layers on top of each other (Campbell *et al.*, 2012; Gibson *et al.*, 2015; Gibson, 2006). Metal and plastic printing, because of their compatibility with computer-aided design (CAD), are suitable for the production of parts with final shapes and have been widely used in different manufacturing applications such as medical, architecture, automobile, microfabrication, electronic and aerospace (Yuan *et al.*, 2015; Bourell, 2015; Gibson, 2015; Gibson *et al.*, 2006; 2002; van Oosten *et al.*, 2009; Wegst *et al.*, 2015). Writing in "*Nature Materials and Medicine*", Bank (2015), Wynn and Ramalingam (2012) showed that the functional longevity of spherical implants for different treatments was improved by making them longer. Using AM and changing the design in CAD process can easily obtain this.

Selective laser melting (SLM) is a new approach to metal printing where powder is spread on a platform and selectively melted as a series of thin layers of the powder based on 3D drawings. This process has been carried out in the chamber with a controlled environment regarding pressure, temperature in an atmosphere of inert gas. Owing to various process parameters such as laser spot diameter, layer thickness, hatch spacing, scan speed, build speed and laser power, SLM is a sophisticated fabrication method. Moreover, the high cooling rate has an effect on fabricated samples and leads to an increase in tensile strength and brittleness. Ti and its alloys are well-adjusted biomaterials with good biocompatibility. Therefore, the use of this alloy has extensively grown in recent years. Because of some phenomena such as balling/residual powders, unstable melt pools, porosity, microcracks, low surface quality and shrinkage, post-processing operations such as machining are needed after printing. High brittleness of selective laser melted parts decreases the machinability and surface quality; therefore, to solve this problem, heat treatment (HT) is recommended (Sercombe *et al.*, 2008; Vrancken *et al.*, 2012; Abe *et al.*, 2003; Khorasani *et al.*, 2016, 2015).

Investigation on microstructure evolution during SLM of Ti-6Al-4V showed that because of localized heat and short

The current issue and full text archive of this journal is available on Emerald Insight at: **www.emeraldinsight.com/1355-2546.htm**

Rapid Prototyping Journal
23/2 (2017) 295–304
© Emerald Publishing Limited [ISSN 1355-2546]
[DOI 10.1108/RPJ-02-2016-0022]

The present work was carried out with the support of the Deakin Advanced Characterisation Facility. The authors would like to thank Nima Haghdadi (IFM, DEAKIN) for his help with the EBSD measurements.

Received 11 February 2016
Revised 21 March 2016
Accepted 25 March 2016

interaction time, epitaxial growth occurs, and an intermetallic Ti_3Al was observed during higher laser power and heat. As a consequence of these phenomena, hardness increased from 37 to 57 HRC and tensile strength improved from 0.9 to 1.45 Gpa. Also, the mechanism of irregularities and drop formation is highly related to thermophysical properties of material and process parameters (Murr *et al.*, 2009; Thijs *et al.*, 2010; Yadroitsev *et al.*, 2014). Investigation on SLM of Ti-6Al-7Nb showed that tensile and compressive strength of parts increased while ductility decreased. It was found that as-built samples have α' martensite hardened by dispersive precipitates. Layered manufacturing leads to anisotropy of mechanical properties. Indeed, investigation on the hardness of printed samples showed that if the build direction is perpendicular to longitudinal axis of the printed samples, the hardness was significantly higher than perpendicular build direction to this axis (Chlebus *et al.*, 2011). A mixture of Ti-6Al-4V and 10 Wt.% Mo had β Ti matrix with randomly dispersed pure Mo particle. HT on this alloy showed higher or equal tensile strength compared to conventional wrought samples. Ti-6Al-4V + 10 Mo showed lower Young's modulus and yield strength (73 Gpa and 860 Mpa, respectively) compared to Ti-6Al-4V (113 Gpa and 950 Mpa, respectively). However, the ductility of this alloy notably increased. These behaviours are ascribable to suppressing β transformation to α' martensitic, therefore the β phase is fully retained. Also, adding Mo results in decreasing martensitic starting temperature and decreasing critical cooling rate to retain β phase (Vrancken *et al.*, 2014). The same observation about the formation of α' and enhancement on mechanical properties such as hardness and tensile strength for CP-Ti has been reported (Attar *et al.*, 2014). Optimization of scanning speed and laser power that were reported as the most effective parameters on SLM leads to enhanced mechanical properties of produced TNM-B1 titanium aluminide alloy and increasing density. Also, HT on 3D printed parts for this alloy resulted in 50 and 30 per cent decrement of offset yield and ultimate compressive strengths, respectively, while ultimate compressive strain increased by 180 per cent (Löber *et al.*, 2014). The interaction of high power and low scanning speed leads to apparent residual powders, wormholes, micro-cracks, increasing mush area in solidification, decreasing liquid viscosity and increasing thermal stresses. In contrast, high scanning speed causes unstable melting pool and increasing residual powders and micro-cracks. As a consequence of these scanning situations, lath-shaped α phase coarsened and martensitic α' appears. Also, in the case of high scan speed, partial evaporation, especially for alloying elements, makes chemical gradients onto the melt pool surface. Changing the surface tension coefficient, reversing the melting flow from bottom to the surface of the melting pool, escaping hydrogen and lower porosity are the results of high scan speed. It is reported that SLM has some drawbacks such as porosity and micro-cracks, and to solve these problems and enhancing mechanical properties, drying powder and optimization process parameters has been suggested.

In this paper, to improve mechanical properties of SLM parts to achieve surgery implants standards, different annealing operations in α, β and $\alpha + \beta$ conditions have been applied. Then microstructure, mechanical properties such as tensile strength, elongation, macro-hardness and fractured surfaces are analysed and compared with conventionally wrought samples to present a better explanation on the behaviour of those two materials.

2. Experimental procedure

2.1 Powder material and SLM operation

Granulomorphometry is a significant factor that affects mechanical properties, distortion and irregularities in SLM parts. In this operation, equiaxed Ti-6Al-4V powder [Figure 1(left)] that is made by plasma atomization was used. Based on the F2924 – 14 ASTM standard, dogbone coupons were designed according to E8/E8M Standard (approved July 1, 2013; Published August 2013) for the tensile test. [Figure 1(right)].

SLM 125HL equipped with a YLR-Faser-Laser, with a maximum laser power of 200 W, a minimum spot size of 5 μm and wavelength of 1,070 nm were used in this experiment. To produce fully dense and higher mechanical properties based on the recommended ranges from the literature and original equipment manufacturers (OEMs), meander pattern scanning and process parameters were selected. Process parameters are shown in Table I. Conventionally wrought samples were forged according to ASTM B381.

2.2 Heat treatment

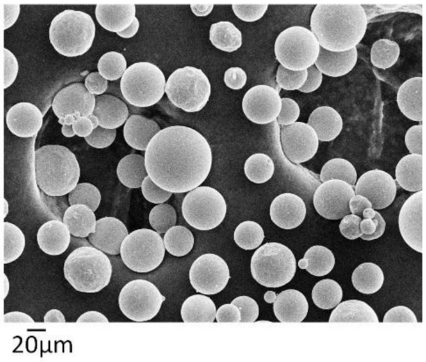
According to ISO 20160, to increase the quality, mechanical properties and material integrity of SLM parts, especially in medical applications, it is necessary to fabricate high-density and homogeneous equiaxial microstructure. Ti-6Al-4V are susceptible to oxidation with oxygen and nitrogen in HT. The results of the oxygen- to nitrogen-rich phase termed alpha-case that affects mechanical properties. To compensate this issue, different HT processes such as stress relief annealing, full annealing or mill annealing, recrystallization annealing and beta annealing have been carried out in vacuum furnace Camco SAN Carlos model G-VAC-2000 with Ar atmosphere and FC. Table II shows the HT conditions.

2.3 Experimental equipment

The microstructures were examined using optical microscopy (OM) and scanning electron microscopy (SEM) equipped with electron back-scattered diffraction (EBSD). For the OM, the samples were mechanically polished and then etched in a solution of 2 per cent HF, 6 per cent HNO_3 and 92 per cent H_2O. For EBSD, the samples were polished by a colloidal silica slurry solution after mechanical polishing. Fracture surfaces were revealed using a field emission Zeiss-SUPRA 55-VP scanning electron microscope. EBSD analysis was performed using an FEG Quanta 3-D FEI SEM under a condition of 20 kV and 4 nA. TSL software was used to acquire and analyse the EBSD data. The step size was assigned to be 0.1 μm. Type of grid was hexagonal and the average confidence index was higher than 0.50 for all the maps.

Tensile tests have been carried out using Instron 8801-100kN machine equipped with hydraulic jaws to determine ultimate tensile strength and percentage of elongation. The Vickers macro-hardness has been carried out using Struers Durajet machine. Alicona Infinite Focus profile-meter with 1-nm resolution and the ability to scan the whole surface was used for scanning fractured surfaces.

Figure 1 (Left) SEM image of powder morphology and size; (Right) Dogbone coupons and printing direction



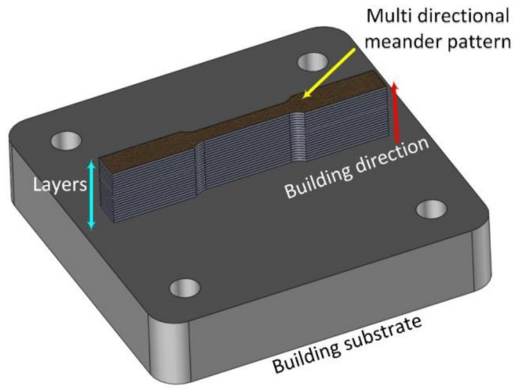


Table I Process parameters for SLM operation in PAC

Process parameters	Value
Build speed	15 cm^3/h
Hatch spacing	75 μm
Laser power	100 W, YLR-Faser-Laser
Laser spot diameter	0.2 mm
Layer thickness	30 μm
Operational beam focuses variable	100 μm
Scan speed	700 mm/s

Table II Various annealing treatments

Annealing	As-built	Stress relief	Mill	Re-crystallization	Beta
Temperature (°C)	20	600	750	925	1,050
Annealing time (min)	–	120	120	120	120
Cooling	–	FC	FC	FC	FC

3. Results and discussion

3.1 Microstructure

3.1.1 Microstructure of conventional wrought samples

The as-received material consisted of primary α and $\alpha + \beta$ phases similar to the result obtained by the literature [Figure 2(a)]. HT at below-*transus* temperature followed by furnace cool does not change the lamellar primary α and has only a small effect on the size of the grains which is shown in Figure 2(b) and (c). Also, in mill annealing depending on previous working different microstructures such as lamellar or equiaxed may be obtained. HT at recrystallization temperature for conventional wrought samples produces different microstructures. Below martensitic temperature, a coarse martensitic $\alpha + \beta$ is reported, while above this temperature, according to lever rule, a duplex microstructure consisting of primary α phase and α coarse lamellae (transferred from β) is found (Murr *et al.*, 2009; Welsch *et al.*, 1993; Jovanović *et al.*, 2006).

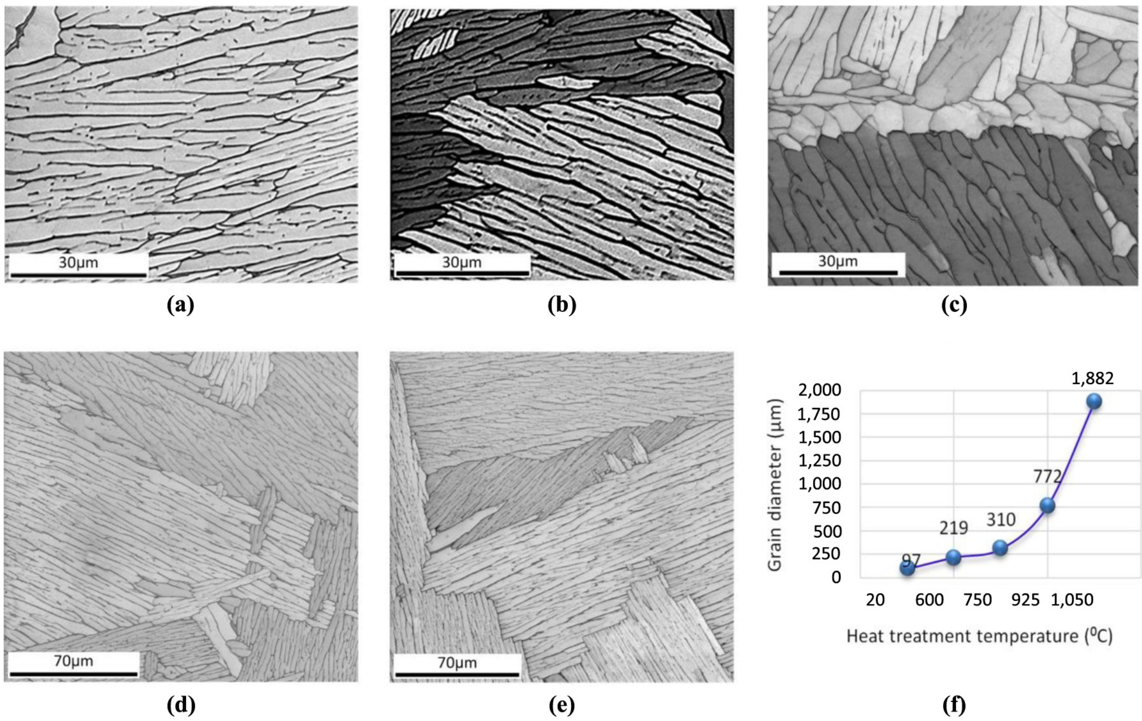
In β annealing of Ti-6Al-4V (1,050°C), a slower cooling rate is a key factor to control diffusion rate between α and β phases because the temperature stays near the β *transus* and α nucleates. As Figure 2(e) shows, the microstructure is lamellar containing broad α phase and fine β lamellae. The size of lamella packets is associated with prior β grain size and cooling rate. The finest lamellae β is found between the fine acicular structure to coarser lamellar which is related to the cooling rate. The degree of microstructural change and homogenization is related to self-diffusion and diffusion of alloying elements. This parameter is a factor of temperature and time. In Ti-6Al-4V, the value of self-diffusion coefficient (D_α Ti) and (D_βTi) are a significant factor in microstructural changes, so according to Table III, homogenization of the microstructure occurs rapidly above β *transus* by increasing grain size and α phase that is observed in Figure 2(d) and (e). In other words, slow-to-moderate cooling from above β *transus* resulted in the nucleation and growth of α phase from β phase grain boundary. Ti-6Al-4V is homogenized by increasing α phase. However, below β-*transus* temperature, microstructural changes have a slow trend (Welsch *et al.*, 1993; ASM Handbooks online, 2010).

3.1.2 Microstructure of SLM samples

Figure 3 shows the microstructure of as-built SLM samples with smaller grain sizes compared to conventional wrought samples (approximately three times smaller) which are associated with a high cooling rate ($10^3 - 10^8$ K s^{-1}) and are in agreement with the literature (Vrancken *et al.*, 2012; Murr *et al.*, 2009; Attar *et al.*, 2014). Furthermore, phase constitution and microstructure are highly associated with the scanning pattern used for the laser. In this work, a meander pattern has been used, and, because of a temperature elevation and high cooling rate, formation of α' martensitic was found to raise tensile strength to 1.35 Gpa and decrease breaking elongation to 3 per cent (Gu *et al.*, 2012; Weingarten *et al.*, 2015; Sieniawski *et al.*, 2013). This microstructure has high fracture toughness, creep resistance and low fatigue crack growth rate.

For SLM samples, α annealing leads to an increase in grain sizes in a fraction of the β phase and relaxation of internal stresses during HT (Baufeld *et al.*, 2011). It is reported that detecting α and α' cannot be easily distinguished by EBSD because of the hexagonal shape of both phases (Baufeld *et al.*, 2011). Annealing of α phase releases high internal stresses that were generated in high working temperature in SLM by diffusion (with a diffusion coefficient in the range of $10^{-19} - 10^{-18}$ m^2/s in 600°C for α and β, respectively), as well as dislocation motion set by Gorny *et al.* (2011). Moreover, in α annealing and furnace cooling (FC) of SLM Ti-6Al-4V similar to conventional wrought samples, primary α is retained

Figure 2 EBSD images for conventional wrought samples



Notes: (a) Untreated; (b) stress relieve annealing; (c) mill annealing; (d) recrystallization annealing (big map); (e) β annealing (big map); (f) grain size

Table III Self -diffusion coefficient for Ti-6Al-4V in different annealing temperatures

Coefficient \ Temperature	600°C	1,000°C
D_αTi (m^2/s)	10^{-19}	10^{-15}
D_βTi (m^2/s)	10^{-18}	10^{-13}

Source: Welsch *et al.* (1993)

and a small value of secondary α grows, so the grain size slightly increased [Figure 3(b) and (c)].

During $\alpha + \beta$ annealing, the microstructure consisting of β and primary α was observed. In contrast to HT with the high cooling rate of this annealing, primary α is not transformed, so the microstructure becomes the combination of acicular, equiaxed and lamellar which are shown in Figure 3(d) and Figure 4(a). This means that after HT, *in-transus* temperature nuclei of globular α-phase appeared. Furthermore, partial re-melting (mush area/heat-affected zone) in SLM results in an anisotropy of mechanical properties and increasing globular α-phase and increasing uniformity. Moreover, β transformation is closely associated with the cooling rate. In the case of slow cooling rate (Table IV), β is transferred to secondary α lamellae; therefore, according to Figure 3(d), a combination of equiaxed, lamellar and acicular was observed after $\alpha + \beta$ annealing (Welsch *et al.*, 1993; Jovanović *et al.*, 2006; Sieniawski *et al.*, 2013; Li *et al.*, 2015).

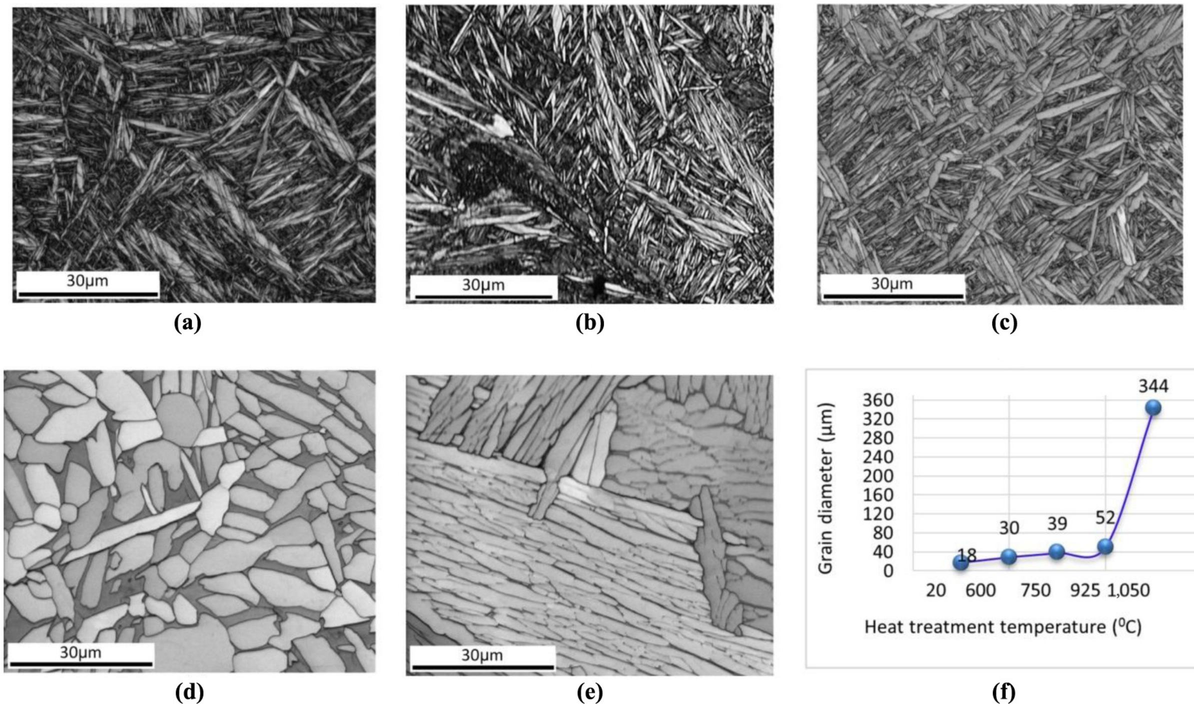
In β annealing of Ti-6Al-4V produced by SLM, the growth of grain size was three times more than as-built samples that are explained by the following reasons. High diffusion rate in 1,050°C and low FC rate (0.1458°C/s) lead to a radical increase in the self-diffusion coefficient, and homogenization of the microstructure consists of primary and secondary α (Yadroitsev *et al.*, 2014; Welsch *et al.*, 1993).

Also, according to Figures 3(d) and 4(b), the volume fraction of β in recrystallization annealing increases to 17.5 per cent and secondary α nucleates on β boundaries, so increasing β boundaries leads to increasing the growth rate of secondary α phase. Moreover, primary α is retained, and the result is a larger grain size consisting of α phase which is shown in Figure 3(e). SLM has been carried out under controlled atmosphere with remaining small value of oxygen and nitrogen; also, Ti has a strong affinity to these elements in local melting zones. It is reported that oxygen and nitrogen contamination with 0.2 and 0.12 per cent, respectively, has been found in the bulk of printed Ti-6Al-4V. This acts as an α stabilizer, with respective higher chemical reactivity between oxygen, nitrogen and Ti. At higher temperature, the value of these two elements and a subsequent α phase increase. Consequently, bigger grains with α phase was observed in β annealing (Gorny *et al.*, 2011; Baufeld *et al.*, 2010; Santos *et al.*, 2002).

3.2 Tensile strength and elongation

The effect of changing annealing temperature on tensile strength and breaking elongation for SLM and conventionally wrought samples was completely different.

Figure 3 EBSD images for SLM samples



Notes: (a) Untreated; (b) stress relieve annealing; (c) mill annealing; (d) recrystallization annealing; (e) β annealing; (f) grain size

Figure 4 (a) EBSD images of recrystallization annealing for SLM samples; (b) volume fraction of α and β phases for SLM samples

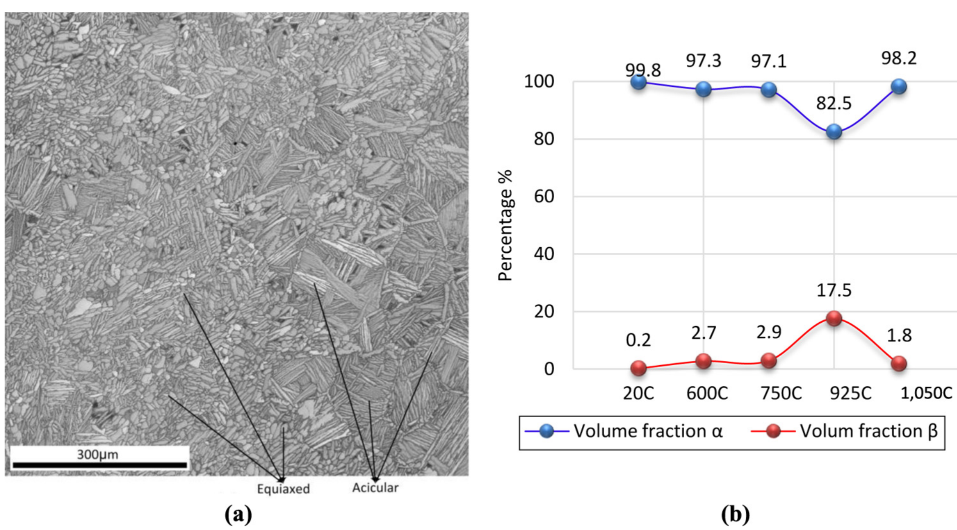


Table IV The effect of cooling rate on phase composition of Ti-6Al-4V

Colling strategy	Cooling rate (°C/s)	Phase composition
Quenching	48-18	$\alpha'(\alpha'')$
Air cool	7-9	$\alpha + \alpha'(\alpha'')$
Air cool	3.5	$\alpha + \alpha''(\alpha'') + \beta$
Furnace cool	0.024-0.4	$\alpha + \beta$

3.2.1 Tensile strength and breaking elongation of SLM samples

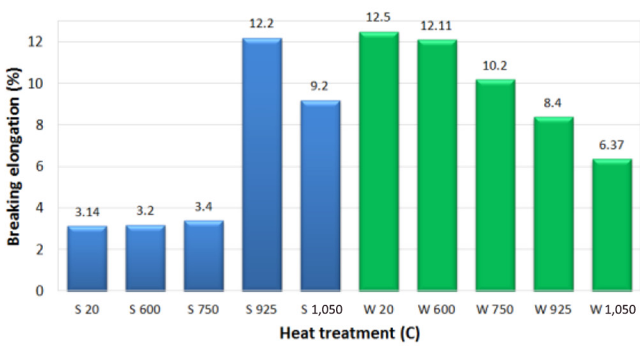
For printed Ti-6Al-4V, high stability of the lamellar microstructure was reported after annealing (Yadroitsev *et al.*, 2014; Welsch *et al.*, 1993). In SLM samples, owing to precipitations in α'-phase, the strength of as-built samples was extremely higher than that of wrought samples (%30). Increasing the holding temperature results in α-phase coarsening and decreasing strength. In stress relieving and mill annealing of SLM samples by increasing temperature and grain size, the value of elongation shows a small increasing trend. Increasing HT temperature in the range of $\alpha + \beta$ results in increasing the percentage of β up to 17.5 per cent and formation of equiaxed microstructure [Figure 4(b)] and a significant increase in elongation. In $\alpha + \beta$ annealing, precipitating of beta on α' leads to a decrease in tensile strength (Yadroitsev *et al.*, 2014). In this HT, because of low cooling rate, primary α retains itself and β is transformed to secondary α. Therefore, the microstructure is the mix of lamellar, equiaxed and acicular. Formation of α-globular can also be related to partial re-melting in the overlap of two hatches. Also, the formation of β is a direct function of low cooling rate and, in this case, β is transferred to secondary α lamellae and primary acicular α is changed to equiaxed α. This microstructure is extremely ductile compared to an acicular structure (Welsch *et al.*, 1993; Jovanović *et al.*, 2006; Sieniawski *et al.*, 2013; Li *et al.*, 2015). The minimum value of tensile strength and breaking elongation according to ISO 5832-3 standards should be 860 Mpa and 10 per cent, respectively, which were satisfied by recrystallization HT (ISO 5832-3, 1996). In β annealing, the value of breaking elongation decreased, which is associated with increasing grain size. Also, a slow-cooling procedure leads to β (that was produced in $\alpha + \beta$ HT) being transferred to secondary α and, by taking retaining primary α into account, the size of α phase increases and a coarse α phase is generated. Moreover, because of higher temperature and slow cooling rate, self-diffusion coefficient increases and a homogenous microstructure consisting of α is observed. In other words, the mentioned phenomena in β HT results in increasing the value of α and grain size and subsequently decreases the value of tensile strength as shown in Figure 5 (Welsch *et al.*, 1993; ASM Handbooks online, 2010). Another reason for decreasing the value of tensile strength is associated with decomposition of the hexagonal α'-martensite, and nucleation of α precipitates at martensite plate boundaries, enrichment of surroundings with V as a β stabilizers which totally lead to the formation of the equilibrium $\alpha + \beta$ phases' mixture (Yadroitsev *et al.*, 2014)

3.2.2 Tensile strength and breaking elongation of wrought samples
As-built commercially wrought Ti-6Al-4V has a tensile strength of 900-950 Mpa depending on previous literature. Lamellar primary α is not changed in below *transus* annealing. However, the volume fraction β increases and, in addition, low cooling rate leads to increasing grain size about three times after mill annealing. Therefore, tensile strength and breaking elongation decrease. In $\alpha + \beta$ annealing, grain size increased seven times, and bigger α laths were observed. This is related to slow cooling rate and retaining primary α and transforming β to secondary α. In Ti-6Al-4V, β phase is stronger yet less ductile, while fine α-phase titanium is more ductile. Hence, increasing the value of α results in decreasing strength and the interaction of increasing coarse α phase and radical growth of grain size results in decreasing breaking elongation (Khorasani *et al.*, 2016; Welsch *et al.*, 1993; Jovanović *et al.*, 2006). In β annealing, owing to higher temperature and lower cooling rate according to Table III, the self-diffusion coefficients (D_αTi) and (D_βTi) increased and a homogenized microstructure was observed [Figure 2(e) and (f)]. These phenomena lead to a radical increase in grain size (about 19 times compared to as-built samples). So, it is concluded that in β annealing, breaking elongation and strength is due to the formation of coarse α, and an extreme increase in grain size decreased (Welsch *et al.*, 1993; ASM Handbooks online, 2010).

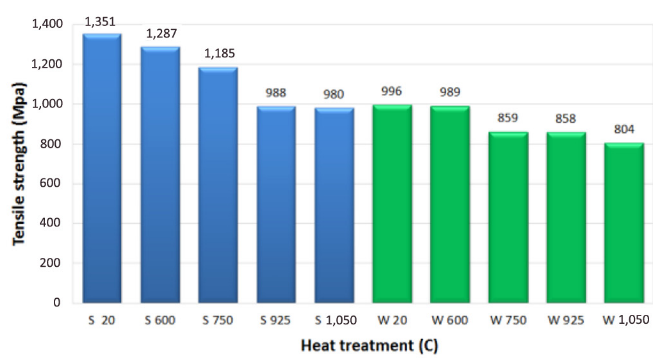
3.3 Macro-hardness

The Vickers hardness measurement on the sample part produced also illustrated that hardness was highly related to density. The use of a laser to form the deposited layers, the subsequent heat-affected zones and temperature distribution determine the degree of martensite decomposition which affect the value of hardness in SLM samples. The hardness of as-built Ti-6Al-4V SLM alloy varies from 350-500 HV (Thijs *et al.*, 2010; Chlebus *et al.*, 2011) and has good agreement with the value of 505 HV obtained in this experiment. The hardness of as-built SLM prototypes was 100 HV higher than wrought material that is related to high cooling rate ($10^3 - 10^8$ K s^{-1}) and precipitation of α' martensitic. According to Figure 6, the value of hardness for SLM samples after α annealing had a decreasing trend. This can be related to increasing the grain size and coarsening of α-phase which is related to HT time and slow cooling rate (Welsch *et al.*, 1993). Decomposition of the α'-martensite and nucleation of α precipitates in $\alpha + \beta$ annealing is a result of increasing the value of secondary α and, subsequently, the total value of α increased and the grain size raise to 52 μm. Subsequently, the

Figure 5 (a) Breaking elongation for various heat-treated samples; (b) tensile strength for different heat-treated samples (S is SLM and W is wrought sample)

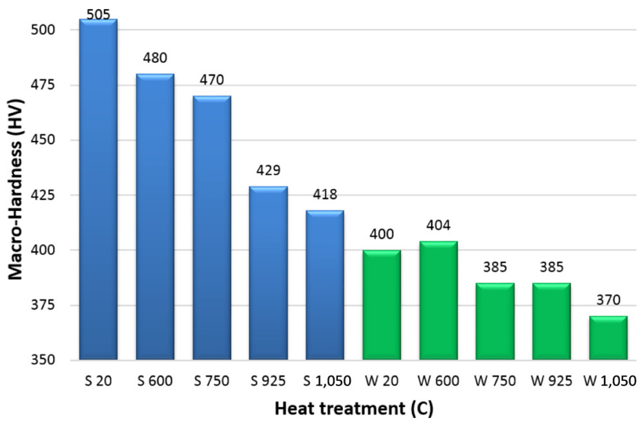


(a)



(b)

Figure 6 Macro-hardness of heat-treated samples



value of hardness decreased to 425 HV. β annealing compares to stress relief and mill annealing has a lower hardness that is related to increasing the grain size and decomposition of α' martensite to secondary α. Indeed, the hardness value of $\alpha + \beta$ annealing was higher than β annealed samples because in $\alpha + \beta$ annealing, the volume fraction of β is higher and grain size is lower. Also, coarsening of α phase in β annealing leads to decreasing the micro-hardness (Murr *et al.*, 2009; Yadroitsev *et al.*, 2014; Chlebus *et al.*, 2011).

The hardness of conventionally wrought Ti-6Al-4V reportedly ranges from 340-410 HV depending on previous stages in the production process (Welsch *et al.*, 1993). Figure 6 shows stress relief annealing due to a small increase in grain size and the volume fraction of α. The value of hardness remains unchanged, but in mill and recrystallization annealing, increasing α and β grain sizes result in decreasing the value of hardness (Chlebus *et al.*, 2011). Because of higher HT temperature, in β annealing higher self-diffusion coefficient, low cooling rate, retaining primary α ductile, nucleation of secondary α coarse and an extreme increase in the grain size, the value of hardness was reduced to 370 HV.

The results of the hardness measurement are in an acceptable agreement with mechanical properties and microstructural observations in this investigation.

3.4 Fractured surfaces

Materials with BCC and HCP crystal structure fractured by cleavage mechanism is shown for as-built and β annealed SLM samples in Figure 7. As can be seen, brittle cracking was affected by pores which are common in this AM process. Lower roughness deviation for as-built samples [Figure 7(a)] was observed in comparison with β annealed samples [Figure 7(c)] that shows increasing ductility and is proved in Figure 5(a). Figure 7(d) shows that smaller micro-pores were observed in β annealed samples that can be related to partial melting on 1,050°C for β annealed samples. Also, river marking cleavage was observed in SLM samples that is a step between crack and cleavage and shows the direction of fracture (Chlebus *et al.*, 2011; Joshi, 2006; Kerr, 1985). The intergranular fracture was observed in printed samples [Figures 7(b) and 7(d)] attributable to weak, brittle grain boundary phases or environmental factors such as hydrogen damage. This fracture surface has a rock candy shape and can be related to micro-void coalescences (Weingarten *et al.*, 2015; Joshi, 2006; Kerr, 1985).

Figure 7 (a) Optical fractography (whole surface scan) of SLM 20°C; (b) SEM image of SLM 20°C; (c) optical fractography (whole surface scan) of SLM 1,050°C; (d) SEM image of SLM 1,050°C

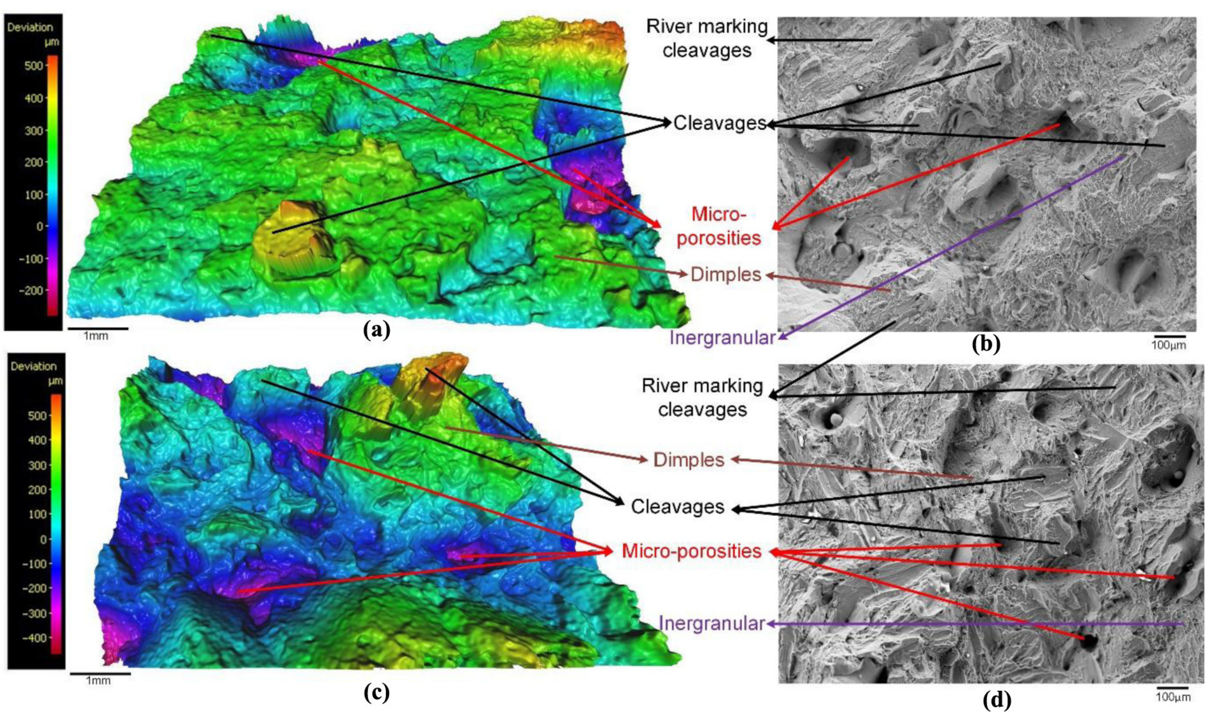
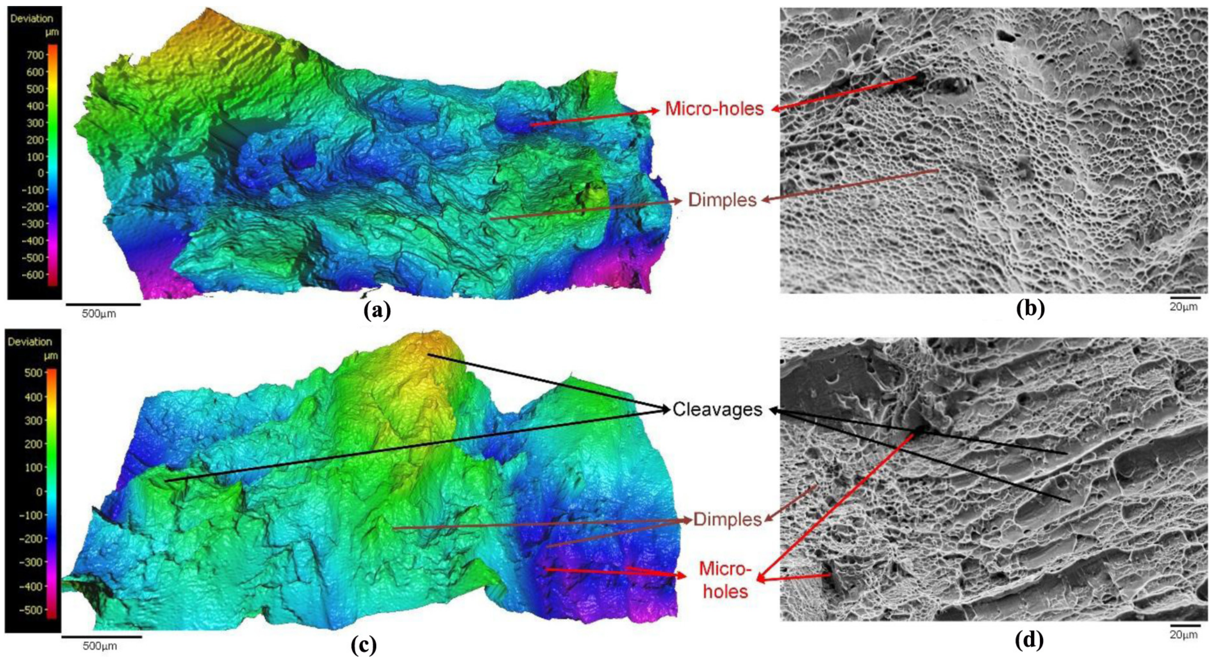


Figure 8 (a) Optical fractography (whole surface scan) of wrought sample 20°C; (b) SEM image of wrought sample 20°C; (c) optical fractography (whole surface scan) of wrought sample 1,050°C; (d) SEM image of wrought sample 1,050°C



Under continually increasing loads, micro-void coalescence is the main reason for fracture. Figure 8 shows the morphology and roughness of fracture surfaces for wrought samples. The as-received samples [Figure 8(a) and (b)] showed characteristic large dimples because of widely spaced nucleation sites. The elongation of this sample was measured at 12.5 per cent, which was the most ductile tested materials in this experiment. The fracture surface of β annealed samples consisted of dimples, micro-holes and a few cleavage-like facets, and the surface roughness was reduced. The cleavages and micro-holes in the β annealed samples can be defined by evaporation of elements such as hydrogen in higher than transformation temperature (Joshi, 2006; Kerr, 1985). Moreover, Figure 8(a) and (c) shows optically filtered images (for whole fractured surface) for as-received and β annealed samples. The scale bars illustrate that β annealed samples had a lower surface roughness, showing more brittle fracture and is proved in Figure 5(a). Finally, comparison of Figures 7(a) and (b), 8(a) and (b) shows that deviation of the fractured surface for wrought samples were significantly higher than those for SLM samples.

4. Conclusion

In this study, based on F 2924-14 ASTM standard, SLM and conventionally wrought samples were fabricated. The effect of different annealing HT such as stress relieve, mill, $\alpha + \beta$ and β annealing on microstructural evolution, mechanical properties, macro-hardness and fracture surfaces has been investigated. The results showed that for SLM parts, by increasing HT temperature, strength decreased while ductility increased, especially after $\alpha + \beta$ annealing, which is sufficient to satisfy medical standards. In conventional wrought samples, by increasing HT temperature, strength and ductility were decreased, and fractography showed the reduction of ductility in β annealing. Furthermore, by increasing HT temperature, the hardness for both printed and wrought samples decreased.

Future research will be directed toward investigation of the effect of different HTs such as aging and various scanning patterns on mechanical properties and microstructural evolution of printed samples.

References

Abe, F., Santos, E.C., Abe, F. and Shiomi, M. (2003), "Influence of forming conditions on the titanium model in rapid prototyping with the selective laser melting process", *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, Vol. 217 No. 1, pp. 119-126.

ASM Handbooks online (2010), *ASM Handbooks Online Volume 2, Properties and Selection: Nonferrous and Special Purpose Materials, Titanium and Titanium Alloy Castings Product Application*, ASTM International, Nasadaq.

Attar, H., Calin, M., Zhang, L.C. and Eckert, J. (2014), "Manufacture by selective laser melting and mechanical behavior of commercially pure titanium", *Materials Science and Engineering: A*, Vol. 593, pp. 170-177.

Bank, R.A. (2015), "Implanted materials: larger is stealthier", *Nature Materials*, Vol. 14 No. 6, pp. 558-559.

Baufeld, B., Brandl, E. and van der Biest, O. (2011), "Wire based additive layer manufacturing: comparison of microstructure and mechanical properties of Ti-6Al-4V components fabricated by laser-beam deposition and shaped metal deposition", *Journal of Materials Processing Technology*, Vol. 211 No. 6, pp. 1146-1158.

Baufeld, B., der Biest, O.V. and Gault, R. (2010), "Additive manufacturing of Ti-6Al-4V components by shaped metal deposition: microstructure and mechanical properties", *Materials & Design*, Vol. 31 No. 1, pp. S106-S111.

Bourell, D.L. (2015), "Perspectives on additive manufacturing", *Annual Review of Materials Research*, Vol. 46 No. 1.

Campbell, I., Bourell, D. and Gibson, I. (2012), "Additive manufacturing: rapid prototyping comes of age", *Rapid Prototyping Journal*, Vol. 18 No. 4, pp. 255-258.

Chlebus, E., KuŸnicka, B., Kurzynowski, T. and Dybała, B. (2011), "Microstructure and mechanical behaviour of Ti-6Al-7Nb alloy produced by selective laser melting", *Materials Characterization*, Vol. 62 No. 5, pp. 488-495.

Gibson, I. (2006), "Rapid prototyping: from product development to medicine and beyond", *Virtual and Physical Prototyping*, Vol. 1 No. 1, pp. 31-42.

Gibson, I. (2015), *Micro Prototyping and Fabrication in Manufacturing, in Handbook of Manufacturing Engineering and Technology*, Springer, New York, NY, pp. 2551-2566.

Gibson, I., Kvan, T. and Wai Ming, L. (2002), "Rapid prototyping for architectural models", *Rapid Prototyping Journal*, Vol. 8 No. 2, pp. 91-95.

Gibson, I., Rosen, D.W. and Stucker, B. (2015), *Additive Manufacturing Technologies*, Springer, New York, NY.

Gibson, I., Cheung, L.K., Chow, S.P., Cheung, W.L., Beh, S.L., Savalani, M. and Lee, S.H. (2006), "The use of rapid prototyping to assist medical applications", *Rapid Prototyping Journal*, Vol. 12 No. 1, pp. 53-58.

Gorny, B., Niendorf, T., Lackmann, J. and Maier, H.J. (2011), "In situ characterization of the deformation and failure behavior of non-stochastic porous structures processed by selective laser melting", *Materials Science and Engineering: A*, Vol. 528 No. 27, pp. 7962-7967.

Gu, D., Hagedorn, Y.-C., Meiners, W., Meng Santos Batista, R.J., Wissenbach, K. and Popraweb, R. (2012), "Densification behavior, microstructure evolution, and wear performance of selective laser melting processed commercially pure titanium", *Acta Materialia*, Vol. 60 No. 9, pp. 3849-3860.

ISO 5832-3 (1996), Implants for Surgery – Metallic Materials – Part 3: Wrought Titanium 6-Aluminium 4-Vanadium Alloy, ISO 5832-3, p. 2.

Joshi, V.A. (2006), *Titanium Alloys: An Atlas of Structures and Fracture Features*, CRC Press, Boca Raton, FL.

Jovanović, M., Tadic, S., Zec, S., Mišković, Z. and Bobić, I. (2006), "The effect of annealing temperatures and cooling rates on microstructure and mechanical properties of investment cast Ti-6Al-4V alloy", *Materials & Design*, Vol. 27 No. 3, pp. 192-199.

Kerr, W.R. (1985), "The effect of hydrogen as a temporary alloying element on the microstructure and tensile properties of Ti-6Al-4V", *Metallurgical Transactions A*, Vol. 16 No. 6, pp. 1077-1087.

Khorasani, A.M., Goldberg, M., Doeven, E.H. and Littlefair, G. (2015), "Titanium in biomedical applications – properties and fabrication: a review", *Journal of Biomaterials and Tissue Engineering*, Vol. 5 No. 8, pp. 593-619.

Khorasani, A.M., Gibson, I., Goldberg, M. and Littlefair, G. (2016), "Production of Ti-6Al-4V acetabular shell using selective laser melting: possible limitations in fabrication", *Rapid Prototyping Journal*, Vol. 216.

Li, X., Roberts, M., Liu, Y.J. and Sercombe, T. (2015), "Effect of substrate temperature on the interface bond between support and substrate during selective laser melting of Al–Ni–Y–Co–La metallic glass", *Materials & Design*, Vol. 65, pp. 1-6.

Löber, L., Schimansky, F.P., Kuhn, U. and Eckert, J. (2014), "Selective laser melting of a beta-solidifying TNM-B1 titanium aluminide alloy", *Journal of Materials Processing Technology*, Vol. 214 No. 9, pp. 1852-1860.

Murr, L., Quinones, S.A., Gaytan, S.M., Lopez, M.I., Rodela, A., Martinez, E.Y., Hernandez, D.H., Martinez, E., Medina, F. and Wicker, R.B. (2009), "Microstructure and mechanical behavior of Ti-6Al-4V produced by rapid-layer manufacturing, for biomedical applications", *Journal of The Mechanical Behavior of Biomedical Materials*, Vol. 2 No. 1, pp. 20-32.

Santos, E.C., Abe, F., Kitamura, Y. and Shiomi, M. (2002), "Mechanical properties of pure titanium models processed by selective laser melting", Proceedings of the Solid.

Sercombe, T., Jones, N., Day, R. and Kop, A. (2008), "Heat treatment of Ti-6Al-7Nb components produced by selective laser melting", *Rapid Prototyping Journal*, Vol. 14 No. 5, pp. 300-304.

Sieniawski, J., Ziaja, W., Kubiak, K. and Motyka, M. (2013), "Microstructure and mechanical properties of high strength two-phase titanium alloys", *Titanium Alloys-Advances in Properties Control*, pp. 69-80.

Thijs, L., Verhaeghe, F., Craeghs, T., Van Humbeeck, J. and Kruth, J.P. (2010), "A study of the microstructural evolution during selective laser melting of Ti-6Al-4V", *Acta Materialia*, Vol. 58 No. 9, pp. 3303-3312.

van Oosten, C.L., Bastiaansen, C.W.M. and Broer, D.J. (2009), "Printed artificial cilia from liquid-crystal network actuators modularly driven by light", *Nature Materials*, Vol. 8 No. 8, pp. 677-682.

Vrancken, B., Thijsa, L., Kruthb, J.P. and Van Humbeeck, J. (2012), "Heat treatment of Ti6Al4V produced by selective laser melting: microstructure and mechanical properties", *Journal of Alloys and Compounds*, Vol. 541, pp. 177-185.

Vrancken, B., Thijs, L., Kruth, J.P. and Van Humbeeck, J. (2014), "Microstructure and mechanical properties of a novel β titanium metallic composite by selective laser melting", *Acta Materialia*, Vol. 68, pp. 150-158.

Wegst, U.G.K., Bai, H., Saiz, E., Tomsia, A.P. and Ritchie, R.O. (2015), "Bioinspired structural materials", *Nature Materials*, Vol. 14 No. 1, pp. 23-36.

Weingarten, C., Buchbinder, D., Pirch, N. and Poprawe, R. (2015), "Formation and reduction of hydrogen porosity during selective laser melting of AlSi10Mg", *Journal of Materials Processing Technology*, Vol. 221, pp. 112-120.

Welsch, G., Boyer, R. and Collings, E. (1993), *Materials Properties Handbook: Titanium Alloys*, ASM International, Nasdaq.

Rapid Prototyping Journal
Volume 23 · Number 2 · 2017 · 295–304

Wynn, T.A. and Ramalingam, T.R. (2012), "Mechanisms of fibrosis: therapeutic translation for fibrotic disease", *Nature Medicine*, Vol. 18 No. 7, pp. 1028-1040.

Yadroitsev, I., Krakhmalev, P. and Yadroitsava, I. (2014), "Selective laser melting of Ti6Al4V alloy for biomedical applications: temperature monitoring and microstructural evolution", *Journal of Alloys and Compounds*, Vol. 583, pp. 404-409.

Yuan, M. and Bourell, D.L. (2015), "Optical properties of laser sintered polyamide 12", *Rapid Prototyping Journal*, Vol. 21 No. 4.

Corresponding author

AmirMahyar Khorasani can be contacted at: mahyar@deakin.edu.au